

SURVEY SPARROW – INTERVIEW EXPERIENCE

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Role: Backend Developer

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Round 1: Written Test

The written test consisted of 30 to 40 questions:

- 20 MCQs (1 mark each)
- 20 Coding Questions (5 marks each)
Questions covered Data Structures & Algorithms (DSA) and DBMS.

DBMS:

- Focused on subqueries and joins
- 3 were easy
- 2 required writing nested subqueries and performing joins

DSA:

- Included a variation of the Buy and Sell Stock problem
- JavaScript output-based questions

 *Negative marking was applicable for each wrong answer.*

25 candidates were shortlisted for the next round.

Round 2: Technical Interview 1 (1 Hour)

This round included a mix of backend, frontend, and project-related questions.

- Explained how Node.js works, focusing on the event loop and concurrency
- Asked about scenarios where Node.js is not suitable
- Questions on React:
 - Basic hooks
 - Difference between React and Vanilla JS
 - Difference between Client-side and Server-side scripting
- Security topics:
 - How I store passwords
 - Why I use bcrypt
 - Working of JWT and cookies
 - How misuse is prevented
- Asked about CORS
- Integration of Nodemailer and Google OAuth
- Deep dive into my project:
 - Implementation
 - Scaling and optimization strategies

- Since I had MongoDB in my resume, they asked how to handle operations without joins
 - DBMS-based question: Subquery and join using student and course tables
 - LeetCode DSA question: Total Maximum Frequency
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Round 3: Technical Interview 2 (1 Hour)

Started with a brief self-introduction, then moved on to the following topics:

- Array vs ArrayList
- Binary Search Tree (BST):
 - What issues BST faces and how to fix them
- AVL Tree:
 - What AVL solves and its limitations
- Red-Black Tree (RB Tree)
- React task:
 - Given ~20 lines of code, asked to find the output (useState)
- LeetCode Question:
 - [Ransom Note](#)
 - For two of my friends, they were asked Word Search

DBMS Query Asked:

```
SELECT tweet_bucket, COUNT(*)
FROM (
    SELECT COUNT(*) AS tweet_bucket
    FROM tweet
    WHERE YEAR(tweet_date) = 2022
    GROUP BY user_id
) AS temp
GROUP BY tweet_bucket;
```

Round 4: Technical Interview 3 (45 Minutes)

- Began with a self-introduction
 - Core concepts were tested
 - Given a partially written code; I had to:
 - Find and fix bugs
 - Implement two functions based on requirements
 - This round was completely based on the interviewer's approach
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Round 5: Technical + HR Round (20 Minutes)

- Started with a system design question:
 - "How would you scale this system?"
 - They expected concepts like Message Queues
- Then moved to personal questions:

- “Where do you see yourself in 5 years?”
- “What is your passion?”
- “Do you have any questions for us?”

✅ *If you reach this round, you're likely selected!*

(Some **React-based questions** were asked too—like hooks and differences with vanilla JS—mainly because some of my answers led them into that direction. I couldn't answer all the React questions, but surprisingly, they didn't treat it as a negative.

I feel like each round had **cumulative scoring** with a **threshold**, which is a fair approach and a good thing in the process.)